



**KIT- KALAI GNARKARUNANIDHI
INSTITUTE OF TECHNOLOGY**



**(An Autonomous Institution, Affiliated to Anna University, Chennai)
Coimbatore – 641402.**

**B23ECP603-INNOVATIVE DESIGN PRACTICES
PROJECT REPORT**

On

**Smart Vehicle Authorisation and Motion-Based Safety
Monitoring System Using IoT**

Submitted by

SATHIYA PRIYA S	711523BEC100
SUGANTHAN B	711523BEC110
RANJITH K	711523BEC310

In partial fulfilment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year 2025 -26

BONAFIDE CERTIFICATE

Certified that this Innovative Design Practices Project Report “**Smart Vehicle Authorisation and Motion – Based Safety Monitoring System Using IoT**” is the Bonafide work of “**SATHIYA PRIYA S (711523BEC100), SUGANTHAN B (711523BEC110), RANJITH K (711523BEC310)**” who carried out the project work under my supervision.

SIGNATURE

Dr S. CHINNAPARAJ
PROJECT SUPERVISOR

Associate Professor,
Department of Electronics
and Communication Engineering,
KIT – Kalaignarkarunanidhi Institute
of Technology,
Coimbatore.

SIGNATURE

Dr M. JAYASHEELA
HEAD OF THE DEPARTMENT

Professor,
Department of Electronics and Communication
Engineering,
KIT – Kalaignarkarunanidhi Institute
Of Technology,
Coimbatore.

Submitted for the project viva voce examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

Owing deeply to the supreme, we extend our sincere thanks to God almighty, who has made all things possible.

We express my heartfull gratitude towards our revered Founder Chairman, **Thiru. Pongalur N. Palanisamy** and Vice Chairperson **Ms. Indu Murugesan** for providing us with the necessary infrastructure to undertake this Project Phase II.

We wish to express our sincere gratitude to our beloved CEO, **Dr N. Mohan Das Gandhi**, and esteemed Principal, **Dr K. Ramsamy**, for fostering an environment and providing the facilities to complete this IDP Project work.

We are grateful to **Dr R. Gowrishankar**, Dean Academics, for his valuable provision and reassurance during this IDP Project.

Our gratitude passes on to **Dr M. Jayasheela**, Professor and Head of the ECE Department, for her valuable support and encouragement during this IDP Project.

We would like to express our special thank of gratitude to **Dr S. Chinnaparaj**, Associate Proffessor/ECE, the Project Guide and **Mr. T.Sathishkumar**, Assistant Professor/ ECE Project Coordinator for her invaluable guidance, feedback, constant encouragement, and support that led to the accomplishment of the project.

The acknowledgement would be incomplete without a word of thanks to our parents, faculty members, and friends for their camaraderie and unwavering support throughout our Project.

**SATHIYA PRIYA S
SUGANTHAN B
RANJITH K**

ABSTRACT

The rise in the number of cars on the road has led to several road safety, accident prevention, and monitoring issues being considered. The present study discusses the design and development of an IoT-based intelligent vehicle system that helps increase the safety of drivers. The system designed and implemented in this project is based on the use of various sensors that monitor the driver drowsiness level, the level of alcohol intake, and accident conditions suddenly. Data collected by these sensors is then processed by a microcontroller, which is basically the control unit for the system. After analyzing the data collected, the system is able to determine if any abnormal or risky situation has occurred. Alerts are issued when an abnormality has been identified, and these alerts are then sent to the user through the IoT platform. Use of IoT technology makes communication between the car and remote monitoring systems possible, thereby making it possible to access information anytime anywhere. Furthermore, consideration is made regarding cost-effectiveness, energy efficiency, and simplicity in implementing the designed smart car solution. This will ensure that the proposed technology works for real-world application in contemporary cars. Combining sensing, processing, and communication technologies provides a perfect way of handling problems of safety in the transport sector. Overall, the smart vehicle concept is a reliable and efficient solution to accident prevention and intelligent vehicle monitoring.

PROGRAM OUTCOMES (PO) MAPPING FOR THE PROPOSED SYSTEM

The developed IoT-based smart vehicle system contributes to the attainment of the following Program Outcomes

PO1: Engineering Knowledge

The project utilizes fundamental concepts of electronics, embedded systems, and communication technologies to design and implement a smart vehicle monitoring system.

PO2: Problem Analysis

Real-time issues such as driver drowsiness, alcohol detection, and accident occurrences are identified and analyzed to develop an effective safety solution.

PO3: Design/Development of Solutions

The system is designed to provide reliable detection and alert mechanisms, ensuring user safety while considering societal and environmental aspects.

PO4: Conduct Investigations of Complex Problems

Experimental testing is carried out to analyze sensor performance and validate system functionality under various operating conditions.

PO5: Modern Tool Usage

The implementation involves the use of modern tools such as IoT platforms, microcontrollers, and programming software for efficient system development.

PO6: The Engineer and Society

The project addresses public safety concerns by minimizing accident risks and promoting responsible driving practices.

PO7: Environment and Sustainability

The system contributes to sustainable development by reducing accident-related damage and supporting safer transportation.

PO8: Ethics

Ethical considerations are maintained by ensuring safety, reliability, and responsible usage of the developed system.

PO9: Individual and Team Work

The project is completed through collaborative teamwork and effective coordination among team members.

PO10: Communication

Technical ideas and results are effectively communicated through reports, presentations, and documentation.

PO11: Project Management and Finance

The system is developed within a limited budget, demonstrating proper planning and resource management.

PO12: Life-long Learning

The project enhances knowledge in IoT and embedded systems, encouraging continuous learning and adaptation to new technologies.

PROGRAM SPECIFIC OUTCOMES (PSO) MAPPING FOR THE PROPOSED SYSTEM

The IoT-based smart vehicle system contributes to the attainment of the following Program Specific Outcomes:

PSO1: Analyze and Design Electronic Systems for Signal Processing and Communication Applications

The proposed system involves the design and integration of electronic components such as sensors and microcontrollers to monitor driver conditions and vehicle status. Signal processing techniques are applied to interpret sensor data, enabling effective communication of information through IoT platforms for real-time monitoring and alerts.

PSO2: Identify and Apply Domain Specific Tools for Design, Analysis, Synthesis and Validation of VLSI and Communication Systems

The project utilizes domain-specific tools such as embedded programming environments, IoT platforms, and simulation tools to design, analyze, and validate the system. Communication technologies are effectively implemented to ensure reliable data transmission between the vehicle and remote monitoring systems.

WASHINGTON ACCORD KNOWLEDGE (WK) MAPPING FOR THE PROPOSED SYSTEM

The IoT-based smart vehicle system contributes to the attainment of the following Washington Accord Knowledge (WK) profiles:

WK1: Natural Sciences and Social Sciences Knowledge

The project applies basic principles of physics and electronics along with an understanding of societal safety needs to develop an effective vehicle monitoring system.

WK2: Mathematics and Data Analysis

Mathematical concepts and data analysis techniques are used to process sensor outputs and interpret real-time data for accurate decision-making.

WK3: Engineering Fundamentals

Core engineering fundamentals such as embedded systems, electronic circuits, and communication principles are utilized in system development.

WK4: Engineering Specialist Knowledge

Specialized knowledge in IoT, sensor integration, and wireless communication is applied to design an advanced smart vehicle system.

WK5: Engineering Design and Sustainability

The system is designed with consideration for efficient resource usage, cost-effectiveness, and reduced environmental impact, supporting sustainable practices.

WK6: Engineering Practice

Practical implementation of hardware components and software programming demonstrates real-world engineering practices in system development.

WK7: Role of Engineering in Society

The project addresses public safety concerns by reducing accidents and promoting responsible driving, highlighting the role of engineering in societal well-being.

WK8: Research and Critical Thinking

The system is developed based on analysis of existing solutions and research literature, encouraging critical thinking and innovative problem-solving.

WK9: Ethics and Professional Responsibility

The project follows ethical practices by ensuring safety, reliability, and responsible use of technology, while respecting societal and user needs.

SUSTAINABLE DEVELOPMENT GOALS (SDG) MAPPING FOR THE PROPOSED SYSTEM

The proposed IoT-based smart vehicle system contributes to the following key Sustainable Development Goals:

SDG 3: Good Health and Well-being

The system improves road safety by detecting driver drowsiness, alcohol consumption, and accidents, thereby helping to reduce injuries and fatalities.

SDG 9: Industry, Innovation and Infrastructure

The project utilizes modern technologies such as IoT and embedded systems to develop an innovative and efficient vehicle monitoring solution.

SDG 11: Sustainable Cities and Communities

By enhancing vehicle safety and reducing accident rates, the system supports the development of safer and more sustainable transportation systems.

SDG 13: Climate Action

The system indirectly contributes to environmental protection by promoting safe driving practices, which can reduce fuel wastage and emissions.

TABLE OF CONTENT

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	4
	PO MAPPING	5
	PSO MAPPING	7
	WK MAPPING	8
	SDG MAPPING	10
1	INTRODUCTION	14
	1.1 INTRODUCTION	14
	1.2 MOTIVATION	14
	1.3 OBJECTIVE	15
	1.4 DOMAIN INTRODUCTION	15
	1.5 PROBLEM DEFINITION	16
	1.6 SCOPE OF THE PROJECT	16
2	LITERATURE SURVEY	17
3	METHODOLOGY	20
	3.1 SYSTEM OVERVIEW	20
	3.2 RIDER/MODE SELECTION	21
	3.3 AUTHORIZATION UNIT	21
	3.4 ALCOHOL DETECTION	21
	3.5CENTRAL PROCESSING UNIT	21
	3.6 SPEED & MOTION CONTROL	22
	3.7 IGNITION CONTROL	22
	3.8 ALERT SYSTEM	22
	3.9 IOT CLOUD	22
	3.10 AUTHORIZATION UNIT	22
	3.11 OVERALL SYSTEM FLOW	23
	3.12 ADVANTAGES	23
4	PROPOSED SOLUTION	24
	4.1 INTRO TO SOLUTION	24
	4.2 WORKING PRINCIPLE	24
	4.3 COMPONENTS	25

CHAPTER NO	TITLE	PAGE NO
	4.3.1 ESP32	25
	4.3.2 MQ-3 ALCOHOL SENSOR	26
	4.3.3 MPU6060	27
	4.3.4 RELAY MODULE	28
	4.3.5 LCD DISPLAY	28
	4.3.6 IOT CLOUD	29
	4.3.7 POWER SUPPLY UNIT	30
	4.3.8 PCB	30
	4.3.9 SUPPORTING COMPONENT	30
	4.4 SUMMARY	30
5	IMPLEMENTATION	32
	5.1 OVERVIEW	32
	5.2HARDWARE	32
	5.3 SOFTWARE	33
	5.4 IOT INTEGRATION	34
	5.5 SYSTEM INTEGRATION	34
	5.6 TESTING	34
	5.7 CHALLENGES	35
	5.8 SUMMARY	35
6	RESULTS AND ANALYSIS	36
	6.1 OVERVIEW	36
	6.2 HARDWARE RESULTS	36
	6.3 MODE SELECTION OUTPUT	36
	6.4 AUTHORIZATION OUTPUT	37
	6.5 IOT MONITORING	37
	6.6 PERFORMANCE	38
	6.7 ANALYSIS	39
	6.8 ADAVANTAGES	40
	6.9 LIMITATIONS	40
	6.10 SUMMARY	41
7	CONCLUSION & SCOPE	42
8	REFERENCES	44

LIST OF FIGURES

FIG NO	FIG NAME	PAGE NO
1	SYSTEM WORKFLOW	20
2	ESP32 MICROCONTROLLER	26
3	MQ-3 ALCOHOL SENSOR	26
4	MOU6050 MOTION SENSOR	27
5	RELAY MODULE	28
6	LCD DISPLAY	29
7	HARDWARE IMPLEMENTATION	33
8	DRIVING MODE DISPLAY OUTPUT	36
9	UNAUTHORIZED ACCESS DETECTION	37
10	SYSTEM VERIFICATION	37
11	IOT DASHBOARD OUTPUT	38
12	ACCURACY COMPARISON	39

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Over the past few decades, due to the increased development of transportation infrastructure and the growth in the number of vehicles used, there has been a substantial increase in the incidence of accidents. These accidents can be attributed to several human factors, such as drowsiness, drinking, distracted driving, and slow responses when dealing with emergencies. Although there are basic safety measures implemented in modern automobiles, there are no smart devices which can constantly monitor the activities of drivers and warn them accordingly.

Internet of things (IoT) technology can be considered as one of the most promising inventions that allows communication among devices and facilitate real time monitoring and exchange of data. IoT technology can be used along with embedded and sensor technologies for the development of smart vehicles that can recognize abnormal situations and take appropriate actions at the moment. The suggested smart vehicle based on IoT will monitor various situations of the driver, including his drowsiness and alcohol intoxication. Apart from that, the proposed system will also detect accidents through the use of suitable sensors. Data collected by sensors will then be processed by the microcontroller and sent as an alert by using IoT.

1.2 MOTIVATION

The rationale behind the development of this project stems from the escalating incidence of road accidents and their serious effects on human beings and their possessions. Several accidents result from easily avoidable causes like tiredness of the driver, alcoholism, and delays in taking action. Upon noticing these problems, it is evident that there is a dire necessity for an innovative solution that would be able to observe the actions of the driver and give instant warnings.

The development of technology within IoT and embedded systems offers a fantastic chance to overcome such difficulties. In fact, the concept of applying real-time surveillance and communication to avoid any accidents is one of the main drivers for developing this project. Moreover, the need to positively affect the lives of people and enhance safety criteria on roads motivates me to work on this project. This project also

promotes innovation and engineering skills to be practically applied in addressing real-life problems.

1.3 OBJECTIVE

The main purpose of the project is to build an intelligent and robust vehicle monitoring system using the IoT technology for improved safety and minimizing the chances of accidents. The system has been planned in such a way that it keeps monitoring the behavior of the driver and recognizes the dangerous conditions including drowsiness and alcohol effects with the help of suitable sensors. Another key purpose of the project is to process the information collected from the sensors with the help of a microcontroller and take immediate actions depending upon the sensed conditions.

Along with safety, the other important aspects that need to be taken care of include making sure that the solution is economical and energy efficient, thereby easy to implement in different kinds of automobiles. The goal of the system is not only limited to detection of faults but also prevention of problems by way of warning the drivers about any possible problems that might take place. In addition to detecting and preventing problems, another important feature of this project is to ensure awareness among motorists regarding how to drive safely.

1.4 DOMAIN INTRODUCTION

This work has been undertaken within the field of IoT and embedded systems, which are one of the most crucial domains of engineering today. The concept of IoT is based on creating a network of interconnected devices that interact with each other via the Internet in order to provide real-time data exchange, while embedded systems refer to the process of combining software and hardware in order to fulfill particular functions.

In the current project, the use of sensors is important in determining the driver conditions like being drowsy and having alcohol in one's body, among others. In addition, it helps in identifying whether there is an accident occurring. In turn, the microcontroller is the processing unit that determines what action to take depending on certain predetermined conditions. The use of the IoT technology allows for remote monitoring and immediate alerting. Users can have access to information wherever they are. Through IoT and embedded system, a very efficient technology has been developed. The field can be used in future improvements like the use of AI technology.

1.5 PROBLEM DEFINITION

The problem of road accidents remains one of the biggest challenges worldwide, with most of the accidents resulting from human faults like tiredness, drinking, and negligence. Current automobile safety systems are inadequate for providing constant monitoring and feedback on the mentioned factors. Therefore, most of the accidents happen due to delayed reactions and a lack of proactive actions. There is definitely an absence of an integrated solution that could detect the risk factors and act instantly.

The issue that is being addressed through this project is the lack of an intelligent, real-time monitoring system that can help in preventing accidents and ensure that drivers remain safe from any harm. The solution involves the use of various sensors for detecting any abnormal situations, followed by the processing of information on the microcontroller and sending out alerts through the Internet of Things. Through such a solution, accident rates could be lowered, and quicker responses would be possible.

1.6 SCOPE OF THE PROJECT

The scope of this project revolves around increasing vehicle safety by deploying an Internet-of-Things (IoT)-enabled intelligent monitoring system. This monitoring system can detect drowsiness among drivers, intoxication, and even accidents and notify the user of the detected events in real time. The monitoring system is deployable in all kinds of vehicles such as private cars, commercial transport, and even public transport systems.

Besides the existing functionality of the system, the future development of the technology can involve the integration of more innovative solutions such as GPS, mobile application development, and cloud computing data analysis. Other possible innovations that can enhance the technology include the implementation of artificial intelligence and machine learning algorithms, which can increase the accuracy and efficiency of identifying hazards. As far as opportunities for future development are concerned, the project can be used as a foundation for creating smart transportation and smart city solutions.

CHAPTER 2

LITERATURE SURVEY

Due to fast-paced advancements in IoT and embedded devices, intelligent approaches for vehicle monitoring and security have been developed. There exist multiple systems incorporating various sensor types and communication systems along with other components to increase road safety and prevent accidents. The following is a brief review of previously introduced smart car systems.

The research carried out in [1] is aimed at developing an IoT-based intelligent vehicle monitoring system that would ensure safety and provide real-time monitoring services. The system makes use of several sensors that would measure different variables such as the status of the vehicle and the activities of the driver. The acquired data is analyzed by a microcontroller and then communicated wirelessly to a distant user interface. This system ensures constant monitoring and can generate alarms in case any irregularities are identified. It demonstrates the significance of timely communication of data to prevent accidents. However, the system is limited to the monitoring process and lacks any preventive measures such as driver fatigue and alcohol detection mechanisms.

The second paper [2] describes an IoT-based cloud-enabled vehicle monitoring system. This system is designed to improve both accessibility and scalability. Real-time data are gathered using sensors and uploaded to cloud platforms where they are stored and analyzed. The users can remotely monitor the data and get notifications whenever abnormal situations arise. Cloud technology makes the management of data efficient and allows scaling up. The results obtained in the paper clearly show the applicability of cloud systems in enhancing efficiency during the monitoring process. Nevertheless, this system is highly dependent on internet connections, and this could limit its performance in places with weak network coverage.

This study focuses on [3] which talks about the use of sensors to determine sudden alterations in the movement of the vehicle. The system makes use of certain criteria like acceleration and impact force to detect the occurrence of any collision. When an accident takes place, automatic alerts are made to the emergency contacts, thus leading to the provision of timely help. It has been found that this approach is quite effective in reducing the delay time associated with the provision of emergency services. The study

shows the significance of automated systems in making roads safer. But, this system does not contain any provisions for accident prevention.

In [4], an all-in-one smart vehicle system has been proposed that incorporates various features such as monitoring the drivers, detecting any accident and monitoring environment. Various sensors along with embedded systems have been used to acquire and process the real-time data. Integration of multiple functionalities within a single system ensures enhanced safety and efficiency of the system. This system also includes Internet-of-Things (IoT) technology to communicate information to the users. Although this methodology ensures better performance of the system, but at the same time makes the system more complex and expensive to implement.

The study in [5] is concerned with continuous monitoring of vehicles using the Internet of Things (IoT). This technique involves monitoring several aspects concerning vehicle performance, and providing real-time feedback for the benefit of users. One can observe that monitoring is an essential part of this study, as its main goal is to identify problems and ensure greater reliability in operation of the proposed system. Nevertheless, the problem of drivers' safety remains ignored, especially when it comes to alcohol intake and fatigue.

The authors in [6] suggest a model for driver behavior monitoring. The proposed solution aims to identify unsafe driving behaviors by analyzing data obtained from sensors. The system evaluates the activities of drivers and alerts them in case of any unsafe behavior. In such a way, it assists in avoiding accidents caused by human mistakes and encourages the drivers to be aware of their driving behavior. Nevertheless, the lack of integration with the Internet of Things platform restricts the possibility of remotely monitoring the behavior and transmitting alerts to other people.

In [7], the author describes a smart transportation system based on the Internet of Things that processes the data collected through various devices in order to make the transport safer and more efficient. This smart transport system applies advanced data analytics to aid decision-making and enhance the performance of transportation networks. The IoT provides the advantage of real-time communication, but the system consumes lots of computing power and infrastructure resources.

In [8], an IoT based system with GPS tracking and two-factor authentication to ensure increased security in tracking and monitoring of vehicles is proposed. This system ensures real time tracking and secure access of the information. With the help of proper authentication systems, increased security can be achieved. From this literature, it can

be concluded that there is an increased need for improved security and tracking in contemporary systems used in vehicles.

From the literature survey above, it is clear that current systems have been concentrating on providing functionality such as accident detection, vehicle tracking, driver behavior analysis, etc., with IoT and various other advanced technologies. Other supportive literatures [9]–[16] have discussed the role of IoT and embedded systems in enhancing vehicle security and real time communication.

Nevertheless, an integrated system that utilizes several safety features in one system is scarce. Most of the available systems focus on solving specific issues but fail to deliver a comprehensive system. In addition, cloud computing infrastructure, complex system design, and costs play a significant role in affecting practicality of the proposed solutions. Moreover, there has been little focus on integration of preventive measures and reactive ones like driver authentication, alcohol testing, fatigue recognition, and movement sensors.

In light of the above, the proposed system seeks to achieve a combination of different aspects like driver authentication, alcohol testing, fatigue identification, and movement sensors through MPU6050 within a single Internet of Things-based system. Real-time monitoring, generation of alerts, and improved efficiency are among the key benefits that will be offered by the system.

CHAPTER 3

METHODOLOGY

3.1 SYSTEM OVERVIEW

The intended system for smart vehicles based on the Internet of Things (IoT) is aimed at improving the safety level and control over a motorized vehicle by analyzing different factors. This IoT-based approach includes several modules like selecting the mode of ridership, authorization device, alcohol testing, detecting movement, and communication over the Internet of Things. The ESP32 microcontroller works as the primary element that analyzes all the input data provided by different sensors and regulates system output. The entire process is organized sequentially and implies collecting data, analyzing, and taking action to provide safety. The system can restrict access to unauthorized users and prevent riding in case of unsafe conditions.

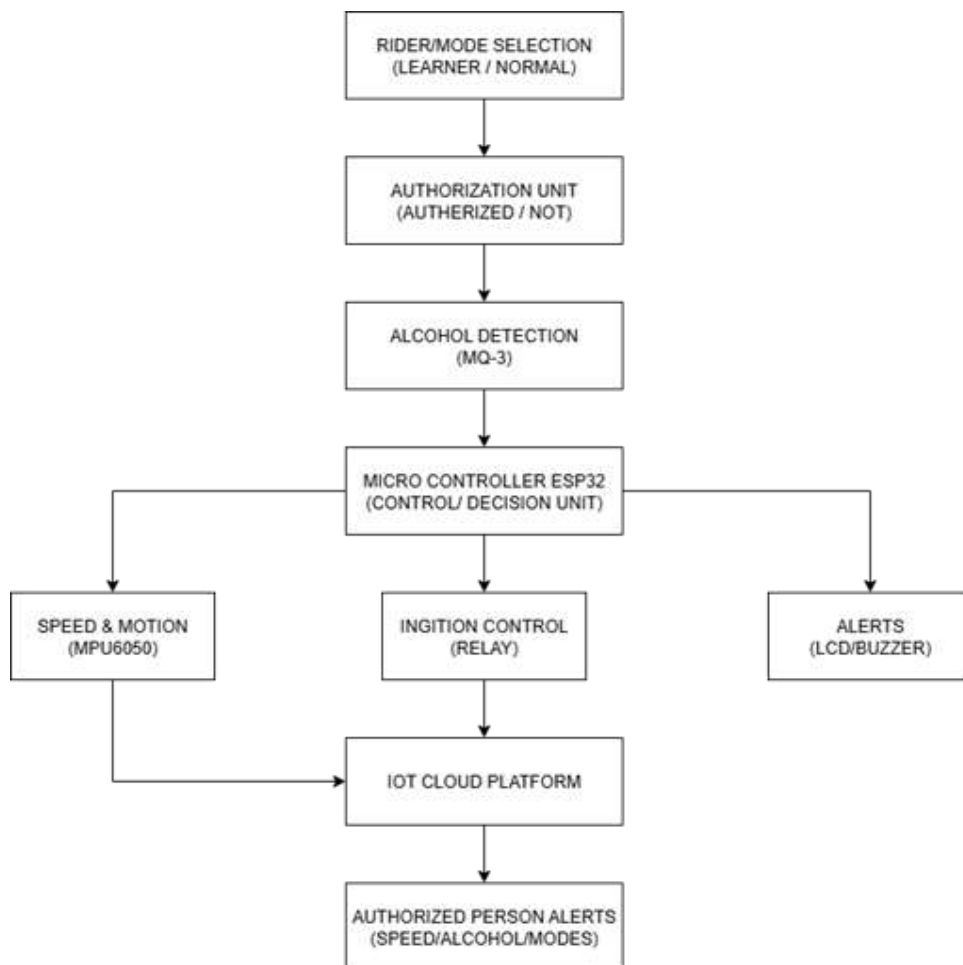


Fig 1. System Workflow

3.2 RIDER/MODE SELECTION

In the first phase of the system, the rider starts by selecting the mode of operation from the various modes available in the vehicle, including the learner mode and normal mode. It is essential because of its ability to manage the vehicle depending on the experience of the rider. In learner mode, speed limits are set to ensure safety, while in normal mode, the vehicle runs as usual. The mode of operation selected is then sent to the microcontroller as an input.

3.3 AUTHORIZATION UNIT

Once the mode selection step is done, the authorization phase will be taken up. In this step, the authorization of the rider will take place. This helps to make sure that the vehicle is driven only by authorized personnel. The methods of authentication in this step may include password authentication, RFID authentication, among others. If the person is an authorized one, then the system permits further processes. Otherwise, the ignition of the vehicle stays locked. This module thus adds a crucial element of security for the system.

3.4 ALCOHOL DETECTION (MQ-3 SENSOR)

After the process of authorization, the system makes use of the MQ-3 alcohol sensor to determine if there was any consumption of alcohol. The MQ-3 alcohol sensor will detect the presence of any alcohol in the breath of the rider and send an analog signal to the microcontroller. This signal will be processed by the ESP32 module and compared with the set threshold values. When the level of alcohol exceeds the threshold level, the system will determine that there is a dangerous situation. In this case, the ignition system will be deactivated, and alarms will be triggered.

3.5 CENTRAL PROCESSING UNIT (ESP32 MICROCONTROLLER)

ESP32 microcontroller is the control center of the entire system. It obtains input from various modules such as mode select module, authentication module, alcohol detection module, and motion detection module. After receiving the data from all modules, the microcontroller performs calculations according to the programming logic and takes action depending on specific criteria. The microcontroller is responsible for controlling the ignition system, initiating alarm systems, and communicating with the IoT infrastructure. The reason why the ESP32 microcontroller is selected is because of its excellent computing performance and suitability for real-time applications.

3.6 SPEED AND MOTION DETECTION (MPU6050)

The MPU6050 sensor is employed for detecting any kind of motion in the car as well as the speed and impact force changes. This is comprised of an accelerometer and gyroscope that sense acceleration and rotation changes. Data from this sensor is constantly fed to the microcontroller for analyzing whether there is any abnormality in the way the car moves; for example, unexpected jerks or speed and collisions. In case of abnormal motion detection, the car will be considered in danger mode.

3.7 IGNITION CONTROL (RELAY MODULE)

The ignition control module controls whether the engine starts or stops depending on decisions made by the system. In this case, a relay is used to regulate the ignition circuitry. When all the requirements including permission and no presence of alcohol in the breath are met, the relay powers up the ignition circuit allowing the engine to start running. In the event there is any danger associated, the relay stops the ignition from powering the engine so that it does not start. This helps avoid any accident arising due to malfunction of the vehicle.

3.8 ALERT SYSTEM (LCD/BUZZER)

The system has an alerting system, which is designed to warn the rider of any dangers. For this purpose, a buzzer is incorporated in the system to issue aural warnings, while an LCD screen displays messages associated with the system, including messages warning against unauthorized access, detection of alcohol, and excessive speed. Such alerts allow the rider to take appropriate action immediately.

3.9 IOT CLOUD COMMUNICATION

The module facilitates seamless interaction between the vehicle and other users. For instance, the ESP32 device transmits information such as speed, status of alcohol, and alerts to the IoT cloud server. Users will thus be able to observe the state of their vehicle even when they are far away. In case of any emergencies, they receive instant notification to act fast.

3.10 AUTHORIZED PERSON ALERT SYSTEM

The purpose of this design is to ensure that appropriate people are notified in case of any anomalies. Alerts associated with speed limits, the presence of alcohol, and mode

of transportation will be sent using the Internet of Things technology. This is aimed at ensuring that the appropriate people are aware of the vehicle's situation at all times.

3.11 OVERALL SYSTEM FLOW

The entire process works in a step-by-step manner. In the first place, the rider determines the method of operation and gets authorized. The next thing is checking the intake of alcohol. When all the parameters match the desired ones, the ignition key turns on, and the vehicle can be driven. When operating, the motion sensor keeps checking whether there is a movement of the vehicle. The microcontroller takes care of the analysis of the collected data. In case of an unusual situation, for instance, the detection of alcohol or movement, the system sends out notifications and even disables the ignition key.

3.12 ADVANTAGES OF THE PROPOSED METHODOLOGY

Advantages of the suggested approach include real-time monitoring, increased safety, and improved security of vehicles. The inclusion of all the components in one module guarantees full functionality. The application of IoT makes remote monitoring possible, as well as quick alerts; the ESP32 allows processing the data effectively. The suggested approach is economical, convenient for implementation, and applicable in practice. The combination of precautionary measures with response mechanisms ensures reliability.

CHAPTER 4

PROPOSED SOLUTION

4.1 INTRODUCTION TO PROPOSED SOLUTION

The proposed system is an IoT-based smart vehicle safety and monitoring solution designed to enhance road safety, prevent unauthorized vehicle usage, and reduce accident risks. The system integrates various modules such as rider authentication, alcohol detection, motion sensing, ignition control, and IoT communication into a single platform. The ESP32 microcontroller acts as the central processing unit, coordinating all system operations. The system continuously monitors driver behavior and vehicle conditions, processes the collected data in real time, and takes appropriate actions based on predefined conditions. By combining multiple safety features into one system, the proposed solution ensures better reliability, improved response time, and efficient vehicle monitoring suitable for real-world applications.

4.2 WORKING PRINCIPLE OF THE PROPOSED SYSTEM

Continuous sensing, data processing and automated decision-making comprise the operation of the proposed system. External to the system/user interaction mechanism is an authorization mechanism for verification of the user's identity before the vehicle can be operated. Once the user has completed the authentication process, the system uses alcohol sensor(s) to check for alcohol in the user's system. If alcohol is detected in the user's system above the established safe level, the vehicle will not start because the ignition has been disabled by the system. If all safety criteria have been met, the ignition system will allow normal vehicle operations. During vehicle operation, the motion sensor (MPU6050) continuously records vehicle motion so that if the vehicle is impacted in a sudden manner or there is an abnormal vehicle motion, it will alert the microcontroller. The microcontroller, in turn, will process the motion data and generate alerts if necessary. Simultaneously, the IoT Cloud will receive the same vehicle motion data from the microcontroller for the purpose of remotely monitoring and issuing alerts. This cycle of operating the vehicle, collecting motion data to monitor the operation of the vehicle, and responding to generate and transmit an alert, create an enhanced level of safety/risk management in operating the vehicle.

4.3 COMPONENTS

The following components are used in the development of the proposed IoT-based smart vehicle system

1. ESP32 Microcontroller
2. MQ-3 Alcohol Sensor
3. MPU6050 Motion Sensor (Accelerometer + Gyroscope)
4. Relay Module (Ignition Control)
5. 16×2 LCD Display
6. Power Supply Unit
7. PCB Board
8. IoT Platform (Blynk)
9. Resistors and Supporting Components

The smart vehicle system that will be developed through an IoT-based method contains different hardware and software components working together for service, monitoring, and controlling purposes, where individual parts of the system contribute to the complete functioning of the system as a whole. Below is a description of the system's hardware used in this project.

4.3.1 ESP32 MICROCONTROLLER

The ESP32 microcontroller serves as the central processing unit for the proposed system and is critical in controlling all the operations. The ESP32 is an efficient and powerful microcontroller that features built-in Wi-Fi and Bluetooth capabilities that make it ideal for use in IoT-based applications. As part of this project, the ESP32 will receive input signals from sensors, such as the MQ-3 alcohol sensor and the MPU6050 motion sensor, and it will follow programmed logic to process those signals to make decisions based on conditions that have been pre-programmed. For example, an authorized driver will be determined to have permission to drive, the presence of alcohol will trigger the alarm, and any abnormal movement will be evaluated to see whether it was caused by one source. The relay, buzzer, or LCD display will be activated based on the results of each of these analyses performed by the ESP32. In addition, it will transmit its real-time data to the IoT cloud platform to allow for remote monitoring. The ability of the ESP32 to handle multiple inputs and outputs at the same time ensures the system will run smoothly. This project has chosen to utilize the ESP32 because it has very fast

processing capabilities, a very low power usage, and exceptional communication protocols.



Fig 2. ESP32 Microcontroller

4.3.2 MQ-3 ALCOHOL SENSOR

The MQ-3 alcohol sensor detects alcohol in the breath of drivers, which makes it an essential part of this vehicle and its overall safety. It consists of a semiconductor sensor that senses alcohol vapour and gives an electrical output corresponding to the alcohol present.



Fig 3. MQ-3 Alcohol Sensor

The MQ3 sensor measures the alcohol vapour and determines a resistive change. The change in resistance gives an analog voltage. The ESP32 microcontroller then reads this

value. The MQ3 sensor is located close to the driver so that the breath alcohol level can be calculated more accurately.

The microcontroller will continuously look for the output of the MQ3 sensor and compare it with a set threshold. If the value exceeds this threshold, the microcontroller will assume that there is a dangerous alcohol concentration in the breath and will disable the ignition of the vehicle through the relay module. The system will generate a warning signal through the buzzer and LCD display. This component is important because it helps keep drunk drivers from being able to drive and therefore increases safety on the road by reducing the number of accidents from drunk driving.

4.3.3 MPU6050 MOTION SENSOR



Fig 4. MPU6050 Sensor

The MPU-6050 is a motion detection sensor composed of both an accelerometer and a gyroscope integrated into one module. The proposed system will utilize this sensor for the purpose of monitoring the motion of a vehicle as well as detecting situations where abnormal movement has occurred; such as sudden jolts of force, collision or travelling above a predetermined speed. The accelerometer measures how fast the vehicle is travelling in a straight line on three different axes while the gyroscope measures how fast the vehicle is rotating around those same axes. The information provided by combining both measurements gives a lot of detail regarding the movement of the vehicle and how well the vehicle is oriented in space. Data collected by the MPU-6050 will be continuously sent to an ESP32 microcontroller where it will be analyzed for any type of unusual patterns that may indicate the vehicle has suddenly changed its motion. If the ESP32 detects a sudden change in motion, the system will classify this event as

potential accident or unsafe driving and will send out alerts for immediate action. This will lead to a quicker response time to these types of events improving safety. The MPU-6050 has become a very popular sensor for this purpose due to its accuracy, small size and reliability, which makes it a great option for motion detection within vehicles.

4.3.4 RELAY MODULE (IGNITION CONTROL)

The purpose of the relay module within this vehicle is to control the ignition system by ensuring the engine will only run in a safe state. The relay serves as an electric switch that opens or closes the circuit based on received commands from the ESP32 microcontroller. In this installation, the relay is connected to the vehicle's ignition system. If all safety constraints are met, which includes successful authorisation to operate the vehicle by the user and that alcohol detection has not been activated, the microcontroller transmits a signal to the relay to close the ignition circuit allowing for the engine to start. Conversely, should any hazards exist, the relay will open the circuit and therefore prohibit the user from starting the vehicle. This automatic functionality of the relay provides a mechanism to prevent vehicles from being operated in unsafe conditions providing a secure and safe environment for all users to operate their vehicles. Furthermore, this relay module is very easy to work with, is very reliable, and will allow for the control of large amperages in high current or low voltage applications.



Fig 5. Relay Module

4.3.5 LCD DISPLAY

A real-time user interface will be provided through the use of an LCD display. The type of messages that will be displayed on the LCD will include but are not limited to: status

of the system, authorization results, alerts when alcohol is detected, and WARNING messages. The LCD has been connected to an ESP32 microcontroller and the microcontroller will control what shows up on the LCD display based on conditions that pertain to the system. This will greatly enhance user interaction with the system by providing clear and immediate information pertaining to the operation of the system. The LCD may display messages to users such as "Authorized," "Alcohol Detected," or "System Ready," making it easier for the user to know the current state of the system and take the appropriate action if necessary. As such, the LCD display is a critical component of the overall design of the system in enhancing both the usability and effectiveness of the system.



Fig 6. LCD Display

4.3.6 IOT CLOUD

In the proposed solution, an IoT cloud platform is used for real-time monitoring and communication. The ESP32 microcontroller communicates with the cloud via built-in Wi-Fi to pass sensor data and state information. Users can view and track vehicle status through mobile apps or web portals that allow them to monitor conditions from afar. Additionally, the system will alert specified users if there are critical events such as accident detection or unsafe driving. The cloud also stores historical data that can be analyzed or referenced later. Connected through an IoT cloud application, the overall system becomes more advanced, effective, and efficient; making it a better fit for current-day usage scenarios.

4.3.7 POWER SUPPLY UNIT

The Power Supply Unit supplies the electric energy needed by all components of the system to operate properly. It provides the necessary stable voltage and current to the microcontroller, sensor, and other modules to allow the system to work correctly. The use of a regulated power supply helps to eliminate fluctuations in the source voltage or current that could potentially cause performance issues for the system. A reliable power supply is needed to ensure that the system operates well continuously and efficiently.

4.3.8 PCB

Wires that connect components together with either a breadboard or printed circuit board (PCBs). Jumper wires can be used to make temporary connections between different prototypes and can also be used to connect parts of an existing circuit. Both jumper wire and PCB connections are used to ensure proper communication of electrical signals between multiple electrical devices and stability of circuits from one end of the circuit to the other.

4.3.9 SUPPORTING COMPONENTS

In order for a circuit to function properly, there are numerous supporting components, including resistors and voltage regulators. Resistors are used to control currents flowing through a circuit; whereas voltage regulators maintain consistent voltage levels in the circuit. Both of these supporting components are vital to the reliability and performance of a system.

4.4 SUMMARY OF PROPOSED SOLUTION

The IoT based smart vehicle system which has been proposed will incorporate many safety and monitoring functions for the purpose of enhancing vehicle security and lowering the risk of accidents. This innovative system utilises both hardware and IoT communication to implement real-time monitoring and provide decision making assistance to users; in essence bringing together all aspects of what makes a vehicle safe into one integrated unit (including sensors, microcontroller, control modules).

The functioning of each component individually contributes to the detection of unsafe conditions (such as the consumption of alcohol, the unauthorised entry into the vehicle and abnormal motion) so that an appropriate response can be taken. The combination of IoT communication and hardware allows for both continuous collection of necessary information and processing of said information to also be capable of generating alerts

when necessary. Utilising an ESP32 chip for the entire system provides for a fast processing time and excellent communication abilities with the IoT architecture. Consequently remote monitoring and immediate notifications regarding incidents (i.e., vehicle break-ins) can be achieved quickly and efficiently via the proposed solution.

The proposed system combines all of these safety features into a single solution to improve upon existing solutions by being both more complete as well as more cost-effective. In total, the proposed IoT based smart vehicle system is a dependable, effective and scalable solution for modern vehicle safety applications that can easily be operated in the production environment.

CHAPTER 5

IMPLEMENTATION AND DEVELOPMENT

5.1 OVERVIEW OF IMPLEMENTATION

The project consists of integrating hardware components (e.g. sensors, actuators, displays), software programming, and IoT communication technologies (e.g. MQTT, Bluetooth, etc.) into an integrated ecosystem. The objective of this is to have the vehicle operate in real-time by collecting data from multiple sensors, processing sensor readings on the microcontroller, and providing appropriate responses based on the readings. The microcontroller at the core of this system will be the ESP321 and will serve as the interconnection for all of the components (including all of the sensors, actuators, etc.) to facilitate a continuous connection for each of the following: hardware integration, software integration, IoT communication, testing and final verification of the entire system. Each of these areas will be key to ensuring there is an overall stable and functional system once everything is integrated and tested.

5.2 HARDWARE IMPLEMENTATION

Assembling and connecting all physical components necessary for the system constitutes what is referred to as hardware implementation of the system. The system's hardware consists of the ESP32 microcontroller as well as sensors like the MQ3 alcohol sensor and the MPU6050 motion sensor that connect to it. The MQ3 alcohol sensor connects to the ESP32 via its analogue input pin to measure alcohol concentration levels, while the MPU6050 connects via I2C communication pins (SDA and SCL) to obtain motion data.

The ESP32 controls the ignition system through a relay module connected to one of its digital output pins. An LCD that displays system status or alerts has been connected to another pin. At least one buzzer has been connected to provide audible alerting. All devices are powered using a regulated power supply for improved reliability across all components during operation. To make sure there are no loose connections due to the wires used to connect devices, jumper wires have been used to connect devices, while either a breadboard or circuit board (PCB) will provide a more stable and reliable means of connecting device components. The hardware design must be precise enough so that data may be accurately acquired and the system will perform as required.

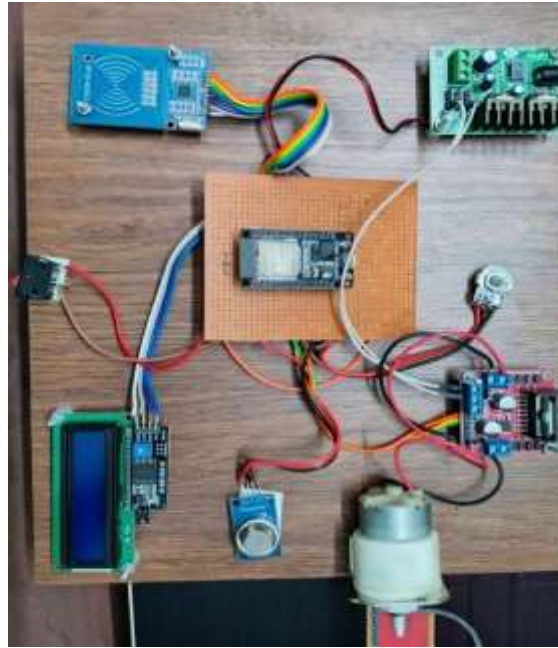


Fig 7. Hardware Implementation of the Proposed System

The above image depicts the hardware design of the suggested smart vehicle system based on IoT technology. In this case, at the centre of the system, is the ESP 32 Micro Controller, which is connected to all other sensors including the alcohol sensor, the motion sensor, the LCD display, the relay module, and the motor unit. Each sensor is connected to each other by jumper connections, and they are mounted onto a PCB board to provide stabilisation during operation. The system is provided power from a regulated power supply, so that consistent performance is provided by the system. This physical representation shows how the proposed smart vehicle system can be built and how it functions under real-world circumstances.

5.3 SOFTWARE DEVELOPMENT

The software for development was completed in the Arduino IDE. The programming language used for programming is an embedded C programming language. The software was responsible for taking readings from the sensor and processing the input and controlling the output to perform actions according to programmed conditions.

The software initializes the sensors, configures the input and output pins, and implements decision-making. The software continuously reads values from the MQ-3 Sensor and compares the values to a predetermined threshold value to determine whether or not the presence of alcohol exists. Furthermore, the software receives data from the MPU6050 Sensor and determines whether or not the data indicates the presence of abnormal conditions (such as sudden impacts or excessive speed).

Conditioning statements are used to control the relay, buzzer, and LCD display. The software also communicates with the IoT system to transmit sensor data. The software is written to continually loop to provide real-time monitoring and respond as quickly as possible. Debugging and testing were performed to ensure an error-free operation.

5.4 IOT INTEGRATION

The IoT integration is very important for the system since it provides remote monitoring and communication. The ESP32 microcontroller takes advantage of its built-in Wi-Fi module and connects itself to any IoT platform, such as Blynk. The data obtained from sensors as well as the status of the whole system is then uploaded into the cloud.

It is possible for users to access the data through their mobile applications and/or web portals, making them able to monitor the condition of the car remotely. If any emergency occurs, such as the detection of alcohol consumption or collision between vehicles, an alert message will be sent to the selected users immediately.

5.5 SYSTEM INTEGRATION

Combining all of the hardware and software components into one complete, working (integrated) system is the objective of system integration. All of the modules have been verified to function independently before integration to guarantee that they communicate properly with one another. The ESP32 will serve as the primary unit that handles inputs and outputs.

Special consideration is given to synchronizing between the sensor(s), microcontroller, and output devices during the integration phase. A continuous loop that collects data from the sensors and processes the data before using it to control the output devices is how the system is intended to run. The components have been integrated together so that they will work together to provide accurate monitoring and an efficient, timely response to any events.

5.6 TESTING AND VALIDATION

Testing and validation processes help determine the efficiency and effectiveness of the system. Various tests are carried out depending on the modules that are involved to verify their functionalities. In the case of alcohol detection, the MQ-3 sensor is exposed to alcohol fumes to see how it works while motion detection is done via sudden movement or impact.

These various tests enable the verification of the system's operation under these conditions. The relay function can be validated to ensure proper functioning of the system when controlling the ignition, and also ensure that there are no errors in the alerts issued using the buzzer and LCD. Finally, IoT communication test validates whether the real-time data transmission takes place.

5.7 CHALLENGES FACED DURING IMPLEMENTATION

While implementing an application or program, several different types of issues may occur. These issues include, but are not limited to: inaccurate or mis-communicated readings because the sensors were improperly calibrated; unstable connections; and delays in communications between devices/controllers. In order for applications and systems to communicate, the sensors must be calibrated correctly, and there should be stable connections with a reliable power supply so that the sensors will produce accurate data. It is also crucial to have a consistent connection to the Internet in order for the Internet of Things to function properly. Properly designing, testing, and optimizing all aspects of the system will mitigate most (if not all) of these types of challenges.

5.8 SUMMARY OF IMPLEMENTATION

In implementing and developing the proposed system, it is necessary to successfully integrate hardware components, software programming, and communication via Internet of Things. The system should be able to perform its operations in real-time, allowing it to monitor accurately and respond promptly to any significant situation. The efficient processing and connectivity capability of ESP32 and the reliable data that the sensors can provide for decision making will help prove the feasibility of the proposed system.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 OVERVIEW OF RESULTS

The design and implementation of an IoT-enabled intelligent vehicle system have been completed and evaluated for their performance and accuracy through various real-time conditions. This has been achieved through the testing of various components such as authorization, alcohol detector, motion sensor, and IoT communication, followed by overall system testing. From the results obtained, it can be noted that the system performs well and can perform monitoring and decision making. It is also possible to generate alerts in a timely manner, thus ensuring the safety of the vehicles.

6.2 HARDWARE IMPLEMENTATION RESULTS

Hardware configuration of the proposed system has been done utilizing the ESP32, sensors, relay module, LCD screen, and many others. Everything has been connected and configured to work on a regulated source of power. The system showed its capability to perform as desired in its operation. This successful configuration proves the possibility of designing this system for practical applications.

6.3 MODE SELECTION AND SYSTEM DISPLAY OUTPUT

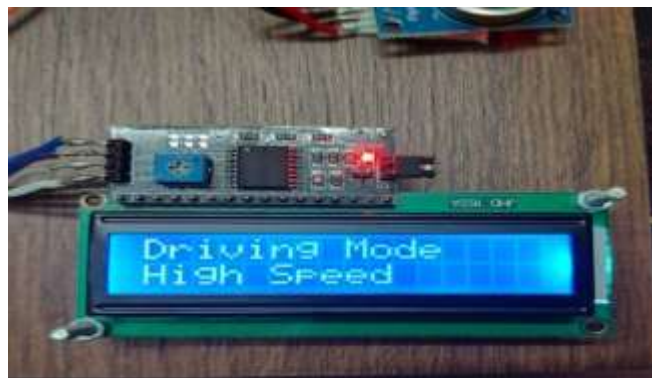


Fig 8. Driving Mode Display Output

The message “Driving Mode High Speed” will be shown on the LCD. It means that the system works in the normal mode where there are no limitations. That means the module for mode selection works fine. The system is capable of performing its functions only if all conditions are met.

6.4 AUTHORIZATION SYSTEM ANALYSIS



Fig 9. Unauthorized Access Detection

In case an unauthorized person tries to operate the car, the system will instantly recognize that the data entered is incorrect, and will display the following message: “Invalid Card Access Denied.” This means that the authorization unit works correctly and does not allow an unauthorized person to use the car.



Fig 10. Authorized Access and System Verification

Once the user has been authenticated, the message “License OK Checking Gas” is shown on the display screen, meaning that the authentication phase has been successfully completed. This phase guarantees that several safety tests will be conducted before the vehicle can be driven, thus improving reliability.

6.5 IOT MONITORING RESULT

The following diagram shows the implementation of an IoT Dashboard through a cloud-based IoT platform such as Blynk. The dashboard indicates the real-time data for gas levels, license, accidents, and speed. The gas level data indicator indicates the current reading obtained from the sensor, whereas the license indicator verifies the user's authentication. Accidents and high-speed indicators provide status information in

accordance with the system's status. Real-time speed values can be seen in the display region.

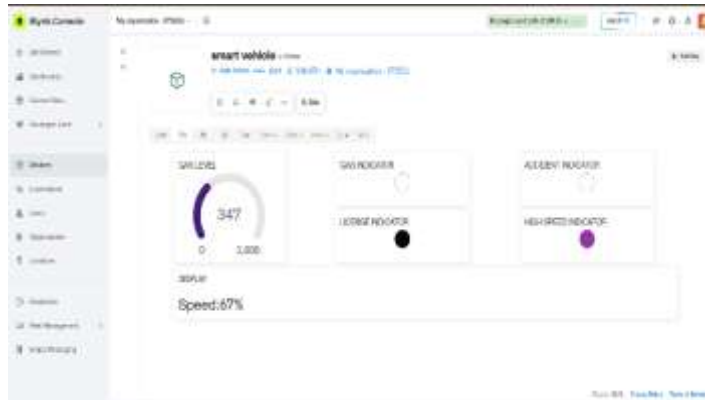


Fig 11. IoT Dashboard Showing Real-Time Vehicle Data

IoT provides remote access to data related to the system, enabling the user to access the information of their vehicle from wherever they are located. Thus, this adds value to the working of the system.

6.6 PERFORMANCE EVALUATION

The performances of the system were analyzed through certain parameters like response time, accuracy, reliability, and communication performance. The ESP32 microcontroller processed sensor data efficiently and produced output instantly. The alcohol sensor (MQ-3) worked properly to detect the presence of alcohol in the air while the MPU6050 sensor worked well to detect any motion or abnormal behavior.

The relay module reacted instantly to the command received and controlled the operation of the ignition process. The LCD and buzzer gave instant feedback to the user. The IoT connection was reliable, and the data was uploaded efficiently to the cloud storage without any delay.

Accuracy is a key metric for evaluating the performance of the suggested system through different measurements: mode selection, authentication, alcohol detection, etc., accident detection, speed control, etc. The graph displays comparisons between the proposals and existing solutions as well as the performance of the planned products. In this case, we found that the new systems have consistently improved upon all measures compared to the old system, due mostly to better sensor integration and lower latency through improved data processing techniques, as well as real-time monitoring using IoT

capabilities. Compared with other systems, the current system provides limited capabilities and an average level of functionality, while the new system provides increased reliability and faster reaction times. The overall data collected shows that the new system operates much more effectively and will deliver more effective results for use in the real world.

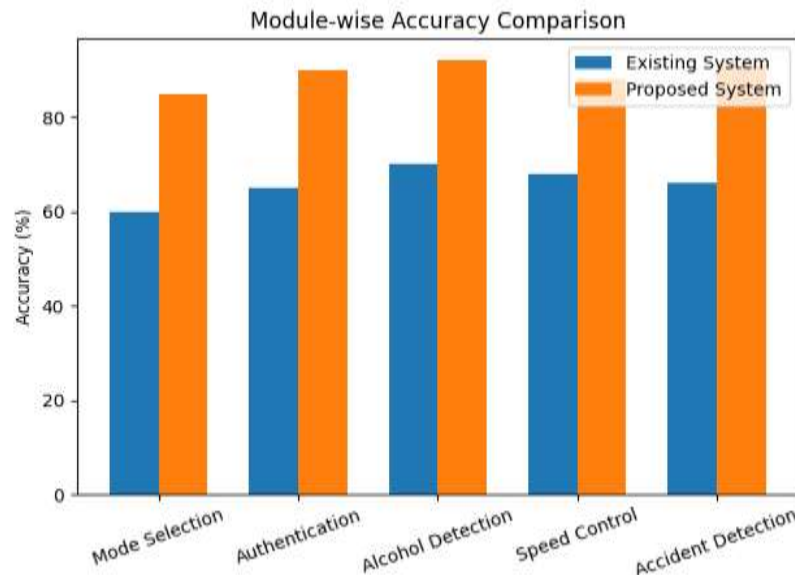


Figure 12. Accuracy Comparison of Existing and Proposed System Across Different Modules

6.7 ANALYSIS OF SYSTEM BEHAVIOR UNDER DIFFERENT CONDITIONS

The innovative smart vehicle system provides dependable operation under a range of circumstances, enabling secure vehicle operation through vehicle entry as well as vehicle control when there are valid users and no alcohol present in the driver. In the event of an unauthorized user attempting to access the vehicle, the system disables the ignition, while providing an alert to the user that access has been denied. If the driver has been detected to have any alcohol present, the system will not permit the vehicle to start, while attaching alerts as required. Once in motion, the motion sensor will continuously track the location of the vehicle while recording any irregular activity (such as a rapid acceleration or deceleration, an abrupt shock to the vehicle, or excessive speed), causing the motion sensor to provide an alert. The system is connected to an Internet of Things (IoT) platform to allow for continual updating of system data from the vehicle to the IoT platform for remote monitoring. Therefore, the system will provide prompt response to all varying scenarios, and thus ensure the vehicle will operate in a secure manner.

S. No	Feature / Scenario	Existing Systems	Proposed System Output	IoT Support	Improvement
1	Driver Authentication	Limited or not included	“Access Denied” / “License OK”	Yes	Secure access control
2	Alcohol Detection	Only detection, no integrated control	Ignition disabled with alert notification	Yes	Prevents drunk driving
3	Motion / Speed Monitoring	Basic or not available	“Driving Mode – High Speed” alert	Yes	Real-time unsafe detection
4	Learner Mode	Not available	“Learner Mode – Safe Driving”	Yes	Controlled driving environment
5	Real-Time Monitoring	Limited or offline systems	Live dashboard with parameters	Yes	Continuous tracking
6	Alert Mechanism	Local alert only	LCD + IoT alert to authorized person	Yes	Remote notification
7	System Integration	Separate modules	Integrated single system	Yes	Efficient design
8	Safety Coverage	Partial safety features	Multi-layer safety (auth + alcohol + motion)	Yes	Comprehensive safety

Table 1. Analytical Comparison of Existing Systems and the Proposed Smart Vehicle Safety System

6.8 ADVANTAGES OBSERVED FROM RESULTS

There is a number of benefits offered by the suggested system that can be clearly seen from the outcomes. Firstly, it allows to monitor everything instantly and to react quickly to an emergency situation. Secondly, it helps to avoid the access of the vehicle by someone who does not have a driver’s license. Moreover, drunk driving is also prevented since it is another reason for accidents.

6.9 LIMITATIONS IDENTIFIED

Despite its effectiveness, there are certain constraints that have been identified during the implementation process. The efficiency of the sensors is dependent on the

environment and should be calibrated to ensure maximum efficiency. In addition, the system relies on an internet connection for the transmission of data using the IoT technology.

6.10 SUMMARY OF RESULTS AND ANALYSIS

The results show that the smart vehicle system based on IoT (Internet of Things) performs well and achieves its goals as planned. This system is able to integrate many safety features and allow real-time monitoring as well as generating alerts for the driver and allowing the driver to communicate with the system remotely. The experimental results indicate that this system is reliable and effective and can be used for many applications. Overall, the proposed solution to improve vehicle safety and reduce the risk of accidents is comprehensive.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

A successful design, development, and implementation of IoT-based Smart Vehicle System that would enhance vehicle safety and real-time monitoring have been achieved. This system combines various elements for safety enhancement including rider mode setting, authentication, alcohol detection, speed detection, and accident detection within a single platform. The ESP32 serves as the main controller that receives sensor input values and implements decision-making algorithms using predefined criteria. The use of IoT platform has enabled constant monitoring and instant alerts for the enhancement of the system performance.

Different tests were performed on the system to verify the efficiency and accuracy of its operations. Test results demonstrate that the system works efficiently in preventing any form of unauthorized access, disabling the operation of the vehicle after detecting alcohol in the user, and initiating alerts whenever the car is running at excessive speed or moving suddenly. This is made possible through the combination of various sensors like MQ-3 and MPU6050 for the detection of unsafe activities and control of the ignition module.

From the results of the performance evaluation, it is clear that the proposed system is more accurate and reliable than other existing systems. The main difference between the existing and proposed systems is that the existing systems concentrate only on one feature while the proposed system integrates several functions. The integration of IoT technology increases the functionality of the system because it allows remote monitoring and access to information in real time. As a result, this helps to minimize response time and take necessary actions when needed.

Another important advantage of the proposed system is that it is cost-effective and relatively simple to implement, which makes it appropriate for use in practical applications. The modular nature of the system makes it easy to upgrade.

7.2 FUTURE SCOPES

However, despite the fact that the system suggested is quite effective in addressing the issues of ensuring safety and monitoring the vehicle, there remains great potential to

make it even better. One of the possibilities includes incorporating the GPS module in the system, thus making it possible to track the exact location of the car at any point in time. This option can be very useful in cases where some kind of accident occurs since it makes it easier and faster to pinpoint the exact location of the vehicle.

Another possibility to make the system better involves creating a special mobile application through which users will be able to interact with the vehicle. By installing the application, users will be able to get instant notifications, monitor vehicle status, and have access to the information collected throughout its use.

Another enhancement will come from the incorporation of Artificial Intelligence and Machine Learning algorithms. These algorithms will help in predicting the driver's behavior and detecting any fatigue and other forms of risk-prone behavior. AI and machine learning algorithms can be incorporated in the system to make predictions and issue warnings based on the current situation.

Some improvements could come from incorporating additional sensors to keep track of the surrounding environment such as temperature and moisture content in the atmosphere. In addition, integrating advanced driver assistance systems in vehicles would improve the functionality of the device. Another development area includes integrating the system with smart cities to enable the management of traffic and emergency situations.

Other improvements would involve enhancing sensor accuracy and minimizing power consumption. Miniaturization and integration with modern cars would improve the system performance. With ongoing developments in IoT and embedded technologies, there is potential to make this device into an intelligent and fully automated car safety system.

CHAPTER 8

REFERENCES

- [1] I. Sapthami, V. V. Narasimha Raju, V. Vaithianathan, P. Chinnasamy, G. Kumaran and R. K. Ayyasamy, "IoT Based Alcohol Detection and Vehicle Control System," *2024 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, Tirunelveli, India, 2024.
- [2] P. P. Hebbal, G. S. Cholaki, M. B, J. R. M, M. N. R and N. G. S, "Alcohol Detection and Life Safety System for Vehicles," *2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, Bangalore, India, 2025.
- [3] H. Akshaya, A. M. Kumar, A. Reddy, D. Dhanalakshmi, S. Hariharan and V. Kukreja, "Ensuring Automobile Safety by Improved Ignition Locking System Through Blood Alcohol Content Measurement and IoT," *2023 7th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, Coimbatore, India, 2023.
- [4] R. S. Vignesh, R. Sankar, A. Balaji, K. S. Kumar, V. Sharmila Bhargavi and R. Anusuya, "IoT Assisted Drunk and Drive People Identification to Avoid Accidents and Ensure Road Safety Measures," *2023 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI)*, Chennai, India, 2023.
- [5] R. Gunawan, R. Nugraha, Kusmadi and Sukadwilinda, "Vehicle Path Tracking Using Inertial Measurement Unit (IMU) Sensors," *2024 18th International Conference on Telecommunication Systems, Services, and Applications (TSSA)*, Bali, Indonesia, 2024.
- [6] A. M. Ali, H. M. Awad and I. K. Abdalgader, "Authenticated Access Control for Vehicle Ignition System by Driver's License and Fingerprint Technology," *2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE)*, Khartoum, Sudan, 2021.
- [7] S. K. T, P. V S, R. T, R. K and S. M, "Smart IoT-based System with Enhanced Safety and Accident Prevention for Two-Wheelers," *2025 6th International Conference on Inventive Research in Computing Applications (ICIRCA)*, Coimbatore, India, 2025.

- [8] R. J. Teradal, S. J. Chougule, S. N. Chopade, A. B. Chougule, V. Dalal and S. S. Atiwadkar, "Comprehensive IoT-Based Vehicle Security: License Verification and Biometric Authentication," *2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*, Goathgaun, Nepal, 2025.
- [9] Z. Ali, S. A. Chaudhry, M. S. Ramzan and F. Al-Turjman, "Securing Smart City Surveillance: A Lightweight Authentication Mechanism for Unmanned Vehicles," in *IEEE Access*, vol. 8, pp. 43711-43724, 2020.
- [10] P. Bagga, A. K. Das, M. Wazid, J. J. P. C. Rodrigues and Y. Park, "Authentication Protocols in Internet of Vehicles: Taxonomy, Analysis, and Challenges," in *IEEE Access*, vol. 8, pp. 54314-54344, 2020.
- [11] G. Pradeepkumar, P. Vijayakumar, N. Chandrasekaran, C. Rohith Bhat, C. Senthilkumar and N. S. Kumar, "Safe Transportation System using IoT based Alcohol Detection," *2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS)*, Madurai, India, 2023.
- [12] V. V, J. S. A, S. S, S. P. R, G. T and A. A, "Enhancing Vehicle Safety: A Comprehensive Approach to Accident Prevention Through Alcohol Detection and Ignition Control," *2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, Chennai, India, 2024.
- [13] M. Muthukumaran, P. R K, P. M, S. S and D. M, "Design and Implementation of IoT-Based Vehicle Over Speed Detection and Alert System on Highways," *2024 International Conference on Sustainable Communication Networks and Application (ICSCNA)*, Theni, India, 2024.
- [14] U. Alvi, M. A. K. Khattak, B. Shabir, A. W. Malik and S. R. Muhammad, "A Comprehensive Study on IoT Based Accident Detection Systems for Smart Vehicles," in *IEEE Access*, vol. 8, pp. 122480-122497, 2020.
- [15] H. C. T., S. Nagaraju, B. N. Varma, K. K. M., M. S. and M. K. V., "IoT based Vehicle Over-Speed Detection and Accident Avoidance System," *2021 International Conference on Disruptive Technologies for Multi-Disciplinary Research and Applications (CENTCON)*, Bengaluru, India, 2021.